SmartAlarm — Detail Design / Detailed Technical Design



Version 1.0 · production architecture design baseline, not deployed · 23/10/2026. This document is used together with API-Contract.json, Event-Schemas.json, Screen-Specifications.md, and Acceptance-Matrix.csv. The attached API contract is a machine-readable core baseline; vendor integrations require their own binding specifications once actual devices are available.

1. Domain data model

IDs use UUIDv7 or an equivalent UUID with uniqueness; external IDs exist only within their source namespace. All tenant-data tables include tenant_id, region_id, created_at in UTC, updated_at, and version bigint. Internal foreign keys use composite tenant_id+id within the same DB; cross-service references use IDs and snapshot versions, without cross-database foreign keys.

Entity	Main fields and constraints	Proposed retention / owner
Tenant	id, home_region, status, policy_version, default_locale; region immutable through normal APIs	Contract end + policy / Tenant
Membership	tenant,user,role,site_scope,valid_from,valid_until,revoked_at; unique tenant,user,role,scope	Audit retains history / IAM
Location	id,parent_id,type,name_i18n,geometry,timezone,version; parent in same tenant; cycle forbidden	Versions retained long term / Location

Entity	Main fields and constraints	Proposed retention / owner
PlanRevision	plan_id,revision,asset_id,scale,exit_graph,reviewer,published_at; unique plan,revision	Based on incident references / Location
Device	id,external_id,connector_id,location_id,cert_thumbprint,capabilities,health; unique tenant,connector,external_id	Device lifetime + audit / Registry
Signal	id,source,event_id,type,observed_at,received_at,schema_version,payload_ref,location_snapshot,mode	30 days hot + archive policy / Ingestion
Incident	id,hazard,severity,state,home_region,writer_epoch,owner,location_ref,version,opened_at,closed_at	Proposed default of 365 days / Incident
IncidentSignal	incident_id,signal_id,relation,actor,reason; unique incident,signal	Based on incident / Incident
SOS	id,client_request_id,person_ref,incident_ref,category,state,location_report,sharing_policy,version	365 days, precise location separate / Assistance
PersonObservation	id,person_ref,incident_ref,status,reporter_type,observed_at,received_at,location_ref	Append-only; precise location 30 days / Assistance
Task	id,incident_ref,unit_ref,state,offer_deadline,version,assignment_epoch,completed_at	365 days / Dispatch
WarningVersion	warning_id,version,content_i18n,area,audience_snapshot,expiry,status,approvals	365 days / Warning
DeliveryAttempt	id,warning_ref,recipient_ref,channel,attempt_no,provider_id,state,receipt_at; effect_key unique	90 days metadata / Delivery
Command	id,target,capability,params_digest,nonce,expires_at,approval,state,device_receipt,writer_epoch	365 days / Command
Mission	id,drone_ref,incident_ref,pilot_ref,plan_version,clearance_ref,state,flight_log_ref	365 days by policy / Command
Evidence	id,tenant,object_key,sha256,mime,size,scan_state,classification,hold_id	90 days, legal hold override / Evidence
Consent	subject,purpose,recipient_scope,fields,policy_version,granted_at,revoked_at	Version history / Privacy
AuditEntry	id,actor,action,object_ref,object_version,reason,trace_id,at,digest,previous_digest	365+ according to policy / Audit
Outbox / Inbox	outbox: event_id,payload,attempt,next_attempt; inbox: consumer,event_id,processed_at	Replay horizon + margin / each service
PhysicalBindingRevision	binding_id,revision,controller_id,regional_model,firmware,gateway_model,gateway_firmware,license_ref,point_map_digest,zone_revision,cause_effect_revision,status,acceptance_ref; unique tenant,binding_id,revision	Referenced configuration history; approved policy / Physical integration
PhysicalPoint	id,binding_id,binding_revision,controller_id,zone_id,raw_address,semantic,access_mode,unit,scale,offset,byte_order,word_order,enum_map,stale_after,feedback_for; unique tenant,binding_id,binding_revision,raw_address	Binding lifetime + incident/test references / Physical integration
PhysicalObservation	id,binding_id,point_id,binding_revision,source_event_id,observed_at,received_at,value,quality,controller_mode,maintenance_state,raw_evidence_ref; append-only	Proposed 30 days hot; archive by incident/test policy / Physical integration
PhysicalCapability	id,binding_id,revision,operation,parameter_schema,allowed_targets,authorization_policy,local_interlocks,command_ttl,retry_class,feedback_point_ids,completion_criterion,unknown_outcome_policy,approval_evidence_ref	Versioned approval history / Command

Entity	Main fields and constraints	Proposed retention / owner
PhysicalCommandDetail	command_id,binding_revision,capability_revision,zone_revision,cause_effect_revision,expected_point_versions,feedback_deadline,physical_state,evidence_refs; one-to-one Command	Inherits Command retention / Command
Impairment	id,scope_refs,functions_affected,reason,owner,planned_until,notifications,compensatory_plan_ref,status,restoration_evidence_refs,version	Site inspection policy; legal hold applies / Physical readiness
PhysicalTestRun	id,binding_revision,test_plan_ref,mode,authorization_ref,impairment_ref,operator,witness,started_at,ended_at,result,evidence_refs	Site inspection/acceptance policy / Physical readiness
CompatibilityAssessment	id,binding_revision,hardware_serials,firmware_versions,license_refs,certificate_refs,fat_run_refs,sat_run_refs,open_defects,signoffs,status	Configuration lifecycle and site policy / Physical readiness
RescueEquipmentMission	id,incident_ref,dispatch_task_ref,equipment_ref,operator_ref,capability_revision,mission_scope,local_console_required,flight_mission_ref,state,limitations,result_evidence_refs	Proposed 365 days; sensitive evidence separately / Rescue equipment operations

Retention values are proposed design defaults, not legal conclusions. Legal holds prevent purging related objects. Deletion retains reference tombstones to preserve audit integrity, without retaining PII beyond its purpose. Search/indexes/backups require deletion propagation and an expiration strategy; deleting from OLTP alone is insufficient.

Indexes and concurrency

Signal unique(tenant_id,source,event_id); index(tenant_id,observed_at DESC), partition by received_at month. Incident index(tenant_id,state,severity,opened_at DESC), spatial index on geometry; partial index for active records. SOS unique(tenant_id,client_request_id), index(tenant_id,incident_ref,state). Task updates use `WHERE version=:expected AND assignment_epoch=:epoch`; affected_rows=0 returns 409. Delivery unique(effect_key); provider callback unique(provider,event_id). Outbox index(status,next_attempt_at) and worker leasing prevent concurrent processing of the same row.

RLS uses tenant context bound from the verified principal within a transaction, not directly trusted headers. Connection pools must reset context. A tenant ID in the URL is insufficient authorization. Region routing checks the membership directory and signed region mapping; wrong-region requests receive safe redirect metadata or 409 REGION_MISMATCH.

Physical storage and revision invariants

Every physical entity also carries the common tenant_id, region_id, created_at, updated_at and version fields above. Use composite references within a service; cross-service references retain the accepted snapshot digest. PhysicalObservation deduplication is unique(tenant_id,binding_id,source_event_id,point_id) when a stable source event ID exists; otherwise the adapter generates a stable receipt ID and documents that it cannot prove source-level deduplication. Index observations by tenant_id,point_id,observed_at and partition by received_at. A point's latest view uses source sequence/time and quality, not arrival order alone.

Binding revisions are immutable after acceptance. Activation atomically changes the active revision only after compatibility acceptance and readiness checks; outstanding command approvals pin their original revisions and are invalidated for dispatch when these no longer match the active binding. physical_state cannot become COMPLETED_VERIFIED without the declared evidence criterion. Impairment restoration and capability re-enable are different transactions; clearing a dashboard alarm cannot restore a device or authorize writes. Snapshot reconciliation compares software maintenance_state with controller mode and reports conflict.

2. API conventions

Base `/v1`; JSON UTF-8, RFC3339 timestamps, cursor pagination. Bearer authorization + verified tenant context; optional X-Tenant-ID is only a selector, checked against token membership by the server. `Idempotency-Key` is mandatory for POST requests creating SOS, sending warnings, offering dispatch tasks, and issuing commands; scope is principal+tenant+route+key, with a request-body hash. Same key with different payload → 409 IDEMPOTENCY_MISMATCH; same key with same payload → previous response. Receipt only after durable commit. Proposed key retention is 7 days for human mutations; source-event deduplication follows an archive replay horizon proposed at a minimum of 90 days, with older replay through a controlled path. Command expiry is independent of deduplication retention.

`If-Match` version/ETag for updates; 412 VERSION_MISMATCH or domain 409 according to API schema. Error envelope: code, message_key, params, trace_id, retryable, field_errors; no exposed stack. 400 malformed, 401 authentication, 403 permission, 404 inaccessible/not found, 409 domain conflict, 412 precondition, 413 payload, 422 validation, 429 quota, 503 unavailable. GET retries are bounded; mutations retry only with an idempotency key.

API	Permission	Request/response and side effect
POST /events	event.ingest + source binding	CloudEvent; 202 durable receipt, not incident success
GET /incidents	incident.read scope	Filters severity/state/site/from; cursor result
POST /incidents	incident.create	hazard,severity,location,signal_refs; 201 aggregate + outbox
POST /incidents/{id}/transitions	incident.transition	expected version,to,reason; validate state guard
POST /sos	sos.create	client_request_id,category,location,note; 201 server receipt
POST /sos/{id}/updates	sos.self_update or sos.manage	append note/location/cancellation request; no deletion
POST /safety-reports	safety.self_report	incident_ref,status,sharing preference; save observation + fan-out event
POST /tasks	task.assign	unit,incident,scope,deadline; offer + workflow timer
POST /tasks/{id}/accept	task.accept + assigned unit	expected version; atomic owner confirmation
POST /warnings	warning.author	localized content,audience,expires; create draft version
POST /warnings/{id}/approvals	warning.approve	version,digest,decision,reason; separate actor guard
POST /warnings/{id}/send	warning.send	approved version; effect key, immutable audience snapshot
GET /deliveries	delivery.read	warning,channel,state; no private address returned without scope
POST /commands	command.request	capability,target,parameters,expiry; 202 requested, not yet actuated
POST /commands/{id}/approve	command.approve	digest,version; publish intent after interlock checks
POST /connectors/{id}/tests	connector.test	fixture_set,manifest_version; sandbox results
POST /connectors/{id}/replay	connector.replay	event IDs,reason; forbid command replay

API	Permission	Request/response and side effect
POST /evidence/upload-sessions	evidence.write	mime,size,classification; short-lived scoped URL
POST /privacy-requests	privacy.self_request	type,scope; identity verification workflow
GET /stream	authenticated subscriptions	event cursor + projection changes; no mutation through socket
POST /physical-bindings	physical.binding.author	controller_id,regional_model,firmware,gateway,point_map,zone_revision,cause_effect_revision; create immutable candidate revision
POST /physical-bindings/{id}/validate	physical.binding.test	revision,fixture_set; schema/point/capability validation, no live actuation
POST /physical-bindings/{id}/activate	physical.binding.activate	expected_version,revision,acceptance_ref; reject absent signoff or incompatible readiness
GET /physical-points	physical.point.read	controller,zone,binding_revision,quality; scoped point projection
GET /physical-points/{id}/observations	physical.point.read	from,to,cursor; values, units, source/receive time, quality and revisions
POST /physical-commands	physical.command.request	capability_id,target,binding_revision,zone_revision,cause_effect_revision,parameters,expires_at; create Command + PhysicalCommandDetail, not actuation
POST /physical-commands/{id}/approvals	physical.command.approve	expected_version,digest,decision; separate actor where policy requires, local guards still apply
POST /physical-commands/{id}/reconciliations	physical.command.reconcile	expected_version,observations,evidence_refs,reason; no automatic resend; preserve UNKNOWN history
POST /impairments	physical.impairment.manage	scope,functions_affected,reason,owner,planned_until,compensatory_plan_ref; create tracked impairment
POST /impairments/{id}/restoration-checks	physical.impairment.restore	expected_version,checklist,evidence_refs,witness; verify before closure; no implicit command
POST /physical-test-runs	physical.test.execute	binding_revision,test_plan_ref,mode,authorization_ref,impairment_ref; approved test lifecycle
POST /physical-test-runs/{id}/evidence	physical.test.record	expected_version,step_id,expected,observed,evidence_refs; append evidence
POST /compatibility-assessments	physical.compatibility.author	binding_revision,hardware,firmware,licenses,certificates,fat_run_refs,sat_run_refs; create review record
POST /compatibility-assessments/{id}/acceptance	physical.compatibility.accept	expected_version,decision,signoffs,open_defects; reject unresolved critical defects
POST /rescue-equipment-missions	rescue.equipment.assign	incident_ref,dispatch_task_ref,equipment_ref,operator_ref,mission_scope,capability_revision; qualified operator workflow, not motion

All physical POST mutations require Idempotency-Key; applicable updates also require expected_version/If-Match. Existing `/commands` routes remain the common command interface; `/physical-commands` is a specialized facade over the same Command aggregate, not a second actuator queue. Unknown/unapproved capabilities return 422 CAPABILITY_NOT_APPROVED; revision drift returns 409 BINDING_REVISION_MISMATCH; active impairment/local

inhibit returns 409 PHYSICAL_INTERLOCK or IMPAIRMENT_ACTIVE; stale prerequisites return 409 PHYSICAL_STATE_STALE. Reconciliation cannot claim a result without supporting evidence. Permissions are site/controller/capability scoped, not tenant-wide by default.

3. Canonical event envelope and semantics

```
{
  "specversion": "1.0",
  "id": "019a0000-0000-7000-8000-000000000041",
  "source": "urn:smartalarm:connector:gw-12",
  "type": "com.smartalarm.signal.fire.observed.v1",
  "subject": "devices/cam-a05-03",
  "time": "2026-10-23T08:40:12Z",
  "datacontenttype": "application/json",
  "tenantid": "tenant-northstar",
  "regionid": "eu-cell-1",
  "mode": "actual",
  "data": {
    "schema_version": "1.0.0",
    "observed_at": "2026-10-23T08:40:11Z",
    "hazard": "fire.possible_flame",
    "location": {"id":"zone-a05-03","method":"registry","version":12},
    "confidence": 0.82,
    "model_version": "flame-v3",
    "evidence_refs": ["evidence-31"]
  }
}
```

The gateway verifies tenantid against credentials; the payload is not the source of truth. source is a registered identity namespace. Event schemas have a version-specific additional-property policy; invalid fields are quarantined with a reason. Detector confidence does not map directly to severity; severity follows policy and impact. observed_at may be delayed or affected by clock errors; server received_at is always present; a skew flag is stored separately. Location geometry uses GeoJSON longitude,latitude; inbound CAP explicitly converts from lat,lon and has polygon tests.

Proposed topics: `sa.<region>.<mode>.signals.v1`, `incidents.v1`, `assistance.v1`, `dispatch.v1`, `warnings.v1`, `delivery.v1`, `commands.v1`, `audit.v1`, `deadletters.v1`. Do not create a topic per user. Broker ACLs are service-specific, with actual/exercise mode isolation. Partition keys use aggregate ID with tenant; telemetry keys use device ID. DLQs contain no secrets; they include an encrypted payload reference, reason, and replay policy.

Physical event contract

Use the same CloudEvents envelope, verified source binding and actual/exercise isolation. Proposed types are `com.smartalarm.physical.point.observed.v1`, `com.smartalarm.physical.command.state_changed.v1`, `com.smartalarm.physical.impairment.changed.v1`, `com.smartalarm.physical.test.completed.v1`, `com.smartalarm.physical.binding.accepted.v1`, and `com.smartalarm.rescue.equipment_mission.updated.v1`. Event data includes schema_version, aggregate_id, aggregate_version, binding_revision where relevant, source provenance and evidence references. The canonical point payload example is a design fixture, not a live observation:

```
{
  "schema_version": "1.0.0",
  "aggregate_id": "point-example",
  "aggregate_version": 1,
  "tenant_id": "tenant-example",
  "site_id": "site-example",
```

```
  "controller_id": "controller-example",
  "binding_id": "binding-example",
  "binding_revision": 1,
  "point_id": "point-example",
  "zone_id": "zone-example",
  "semantic": "pump_running",
  "value": true,
  "unit": "boolean",
  "observed_at": "2026-10-23T08:40:11Z",
  "received_at": "2026-10-23T08:40:12Z",
  "quality": "GOOD",
  "controller_mode": "AUTO",
  "maintenance_state": "IN_SERVICE",
  "cause_effect_revision": 1,
  "source_event_id": "source-example-1",
  "raw_evidence_ref": "evidence-example"
}
```

`pump_running` is not `waterflow_confirmed` and neither means fire extinguished. Units/scaling are defined by the accepted point contract; missing sensors remain UNKNOWN. Command events carry command_id, capability_revision, previous_state, state, controller_receipt_ref, feedback_observation_ids, unknown_reason and source/receive times as applicable. Impairment events carry scope/functions and restoration status; test events carry authorization and evidence references; mission events carry operator and supported-interface mode. Do not publish raw medical notes in equipment events.

4. Sequence: signal → incident → warning

```
sequenceDiagram
  participant E as Edge
  participant I as Ingestion
  participant B as Broker
  participant C as Correlation
  participant S as Incident
  participant W as Workflow
  participant D as Delivery
  E->>I: event + source credential + stable ID
  I->>I: verify / durable store + outbox
  I-->>E: received (not incident confirmed)
  I->>B: signal.accepted (retryable publish)
  B->>C: at-least-once delivery
  C->>S: create candidate (idempotency key)
  S->>S: commit aggregate + outbox
  S->>B: incident.created
  B->>W: start versioned playbook
  W->>W: approved template or human approval
  W->>D: enqueue warning version + audience
  D->>D: channel attempts + feedback tracking
```

If the Incident API times out after commit, the caller retries with the same key and receives the existing incident. If the broker is down, the outbox persists; ingestion still has bounded durable capacity and a degraded state, and does not ACK before commit. If correlation fails, queue-lag alerts and raw-signal triage remain available to operators; manual SOS is not blocked.

For physical alarms, the source is the approved local controller/gateway and preserves alarm/supervisory/trouble distinctions. Local automatic suppression operates on its own approved inputs and cause-and-effect; this cloud sequence only creates the incident and response workflow. Closing the incident does not silence/reset the panel,

stop a fire pump, close a valve, or cancel local protection. Such actions require distinct permitted capabilities and site authority.

5. Sequence: offline SOS and reconciliation

The client creates a UUID request, saves it in an encrypted local outbox, and displays “not sent”. When online, it POSTs with the same request ID; the server commits before returning a receipt. If the response is lost, the client retries and the server returns the existing SOS ID. Only then does the client display received; human ACK arrives as a separate event. Offline location updates always include observed_at; the server retains history without overwriting a newer location simply because an update arrived late. Offline cancellation is a queued request, not a promise to stop dispatch. For a lost device or revoked account, the app purges secure cache according to policy when possible; it does not promise remote wipe while a device remains offline.

6. Workflows and deadlines

Workflow ID = tenant:aggregate:playbook_version. Timers run server-side; client cron polling is not a safety mechanism. Each step has an input schema, activity idempotency key, deadline, retry policy, compensation policy, and manual fallback. Retries use exponential backoff+jitter with maximum attempts/time; business rejections are not retried. An escalation step key of workflow+step+recipient prevents unbounded duplicate sends. Warning expiry cancels unsent jobs; sent jobs require a cancellation/update message and cannot be recalled by deleting DB records.

Situation	Proposed baseline timer	On expiry
P1 incident without operator ACK	60 s	Shift supervisor, then regional backup at 120 s
Unaccepted task offer	60 s	Expire offer; dispatcher selects another unit
Possible fall check-in	Configurable 30 s	Create care request; do not make a diagnosis
Physical command	TTL 30 s or capability-specific	Expire before dispatch; reconcile unknown after dispatch
Webhook delivery	Backoff for up to 24 h or shorter event expiry	DLQ + owner alert

These product defaults require site validation; policy includes min/max and an approver, preventing administrators from arbitrarily changing timeouts during an incident without recording a version.

Physical workflows in baseline 1.0 are WF-28 point/zone onboarding; WF-29 physical supervision; WF-30 authorized actuation, feedback and UNKNOWN reconciliation; WF-31 impairment/test/restoration; WF-32 FAT/SAT compatibility; WF-33 rescue-equipment mission. Their human deadlines and command TTL are capability/site configuration with approvers. No universal software timeout authorizes release, reset, silence or automatic pump stopping. The 30 s command default above is only a proposal and must be overridden or rejected when inconsistent with the approved equipment design.

7. Correlation algorithm and hazard expansion

Exact deduplication precedes fuzzy correlation. Candidate queries use tenant, mode, hazard compatibility, location graph, and event window; scoring uses time distance, same zone, source diversity, and semantic match. Scores are suggestions unless an automation rule is approved. Rule inputs include source confidence/quality; two cameras

using the same model are not counted as two wholly independent pieces of evidence. When several incident candidates exist, the operator chooses; source corrections append correction events instead of editing original events.

The hazard registry contains key, localized labels, severity policy, required fields, playbook binding, sensitivity, default audience, detector schema, and retention class. Add “gas leak”, “industrial spill”, “lone worker”, “missing person”, or “infrastructure outage” through registry/schema/plugins after domain review, without changing hard-coded enums in every service. A new category does not automatically acquire physical-command permissions.

8. Command safety protocol

Requests contain command_id, device_id, capability, canonical params digest, tenant, writer_epoch, nonce, issued_at, expires_at, requester, and approval evidence. Approval signs the digest, so parameter changes invalidate it. The gateway checks credentials, epoch, TTL, capability, zone, local interlocks, and used nonces. Non-idempotent capabilities return retry_class=manual_reconcile; they are not automatically resent when ACK is lost. Local controller results record source time and observed physical state when available. Success requires the capability-specific completion criterion; accepted/executing/unknown must appear separately in the UI.

HA command workers use distributed fencing; failover does not reissue expired commands. Local safety interlocks may reject commands even after cloud approval. Emergency stop appears only when the device supports the correct semantics; a stop-request receipt does not prove the device has stopped. The drone gateway includes flight-stack capability negotiation, altitude frame, link health, pilot override, and lost-link policy; video streaming is separate from the command channel. MAVLink command cancel is not treated as a universal guarantee when unsupported by the flight stack.

Physical command execution sequence

Cloud policy/approval → OT DMZ authenticated conduit → site edge durable command ledger/fencing → approved vendor gateway → local controller → physical device → independent feedback where available. The local protection sequence and manual controls do not depend on this cloud path. The adapter's access profile is observe-only unless exact model/firmware/license and site approval establish a specific write capability; a BACnet WriteProperty advertisement alone is insufficient. Use the restrictions in Physical Device Catalog and Rescue/IoT Device Catalog.

Physical states are DRAFT, PENDING_APPROVAL, AUTHORIZED, SENT, ACCEPTED, EXECUTING, COMPLETED_VERIFIED, REJECTED, EXPIRED, FAILED, UNKNOWN and RECONCILED. Map legacy Command `succeeded` only when its physical completion criterion is verified; retain the richer physical_state in PhysicalCommandDetail. Bind approval to target, capability revision, point/zone/cause-and-effect revisions, parameter digest, TTL and writer epoch. Edge checks freshness, local mode, impairment, allowed operation and feedback availability before sending; the local controller retains final interlocks. After send, a timeout is UNKNOWN unless trustworthy evidence proves failure. Reconciliation appends evidence and a reason; it never silently replays the command.

Wet-pipe thermal sprinkler heads are not generic addressable actuators. Controlled-release assemblies keep their approved local releasing logic; no generic relay bypass, AI-confidence trigger or LLM-generated cause-and-effect is allowed. Approved supervised I/O follows exact electrical, supervision and feedback design. `release_output_active`, valve position, pump running, pressure and waterflow are different observations. Manual-console-only rescue equipment receives a mission assignment, not invented drive/aim/discharge API calls. An unverified remote emergency-stop interface cannot be presented as guaranteed physical stop.

9. Localization vi / en / ja

UI keys use forms such as `incident.status.active`; versioned resource bundles and ICU MessageFormat handle plural/select without concatenated fragments. User locale priority: explicit choice → profile → supported OS locale → tenant default → reviewed English. Language selection does not change region or authorization. Font stack: system UI + Noto Sans when deploying a font bundle; Japanese uses Noto Sans JP/system CJK; no forced uppercase for Japanese; minimum body line height 1.5, with 40% text expansion for testing. Dates/numbers/units follow locale, with timezone shown alongside commands/incidents spanning countries.

Content templates are entities with `template_id`, `version`, `locale`, `text`, `variables_schema`, `reviewer`, `approved_at`, and `voice_asset_ref`. Warning approval references exact locale versions. If an approved translation is missing, clearly show the fallback language and require the publisher to confirm policy; do not generate and automatically publish an AI translation. Do not block all P1 communication when an emergency fallback template is preapproved. Original messages/user chat retain their language; assistive translations are labeled machine translated and do not replace original evidence. TTS voices follow locale and always have transcripts; text remains readable if TTS fails. Critical translations require native-speaker/domain-reviewer checks before release.

Physical terminology must consistently distinguish “request authorized”, “controller accepted”, “physical outcome verified”, “outcome unknown”, “test”, “impaired” and “restored” in vi/en/ja. Preserve manufacturer point labels alongside reviewed localized semantics. Translations never change register semantics, control polarity, parameter units or authority. Speaker TTS/language support is model-specific; use reviewed recordings where the researched payload lacks a verified locale or API.

10. Evidence, chat, and privacy

Baseline upload policy: 20 MB/file for images/documents/short clips; MIME sniffing + extension allowlist + antivirus + decompression limits; quarantine before scanning. Signed upload URLs are scoped to object keys with short expiry; download URLs check policy and TTL, with no public bucket. Video session tokens are incident-scoped, with audited access and region-resident relays. Webhook payloads do not send raw medical notes by default. Chat retention follows incident/social class, with abuse reporting/restricted threads and a shared attachment policy. Safety fan-out snapshots include group membership versions, but revocation is rechecked at send time.

Precise location has field-level protection; generalized site status is the family default. Retention purges include DB, object store, search index, caches, and export links; backups expire and restores reapply tombstones. Conflicts between legal holds and data-subject deletion go to privacy review; requests are not silently ignored. Audit hash chains/tamper-evident storage support modification detection, without claiming absolute immutability if all authority is compromised.

FAT/SAT records include exact serial/firmware/license/configuration, expected and observed point states, timestamps, witnesses, approved test authorization, deviations and signed restoration. Store raw captures separately with hashes and access controls; redact secrets without discarding traceable source references. A product brochure or component certificate cannot set field_accepted. Acceptance references the specific installed binding and approved scope; firmware, point-map or cause-and-effect changes reopen affected review.

11. Frontend runtime and state design

V2 presentation contract and transition tests

`getContent()` selects a renderer by group; `SAEnterprise.render(s,c)` selects composition by screen ID; `SAVisual.scene()` draws schematics using locale and state. A generic `type` property must not determine every layout. `data-go` means navigation and `data-demo` means illustrative interaction; neither is API or actuator success. Existing FRs and `Screen-Registry.json` remain the source contract.

Production applies the following rules before rendering and speech. These are implementation rules, not new API schemas:

Input / transition	Required result and assertion
Heading unknown, floor known	Retain valid plan/landmarks; remove relative arrows and "turn left/right." A QR position fix does not establish heading.
Floor unknown / route withdrawn / evidence stale	Cancel active and queued speech; remove old route overlay and instructions in the same update; retain warning, reported-location entry and assistance. Back/history cannot restore the old route.
Late snapshot older than a withdrawn revision	Reject the obsolete update by revision; HTTP response time does not establish route freshness.
SOS local pending → durable receipt → human ACK	Advance only on evidence for the correct request ID; clicks or hash navigation never create receipts.
Command accepted, readback absent	F11/F12 remain UNKNOWN; no physical retry action appears. Identify missing evidence and reconciliation.
Enter C27 / change language / leave guidance	Cancel old audio; C27 remains silent after return, reload, state changes or new notifications.

Component tests assert text and accessibility properties alongside SVG, rather than image snapshots alone. Integration tests change revisions during speech, interrupt connectivity between send and receipt, delay ACK and assert local override. E2E tests measure focus, targets, reflow, keyboard access and equivalent map text in VI/EN/JA. The current prototype has fixed sample speech, local state and illustrative data; it does not contain production mechanisms proving these assertions.

Server-state caches are keyed by tenant+region+user scope+resource; workspace switching purges or fully partitions them. Client stores hold only drafts/UI preferences and are not the source of incident truth. A server-rendered shell or static app + BFF is selected per deployment; strict CSP, no tokens in localStorage, secure cookies for web/session BFF. Native secure keystores hold tokens, with an encrypted offline database. Websocket sequence gap → snapshot refresh; stale banners show age. Command buttons are disabled when capability/location/policy staleness exceeds thresholds, with a reason and manual channel.

Form pattern: labels always visible; required/error indicators per field; pending submission state; receipt after server ACK; conflicts show server version and draft without automatic overwrite. List pattern: shareable filters without PII in URLs; accessible tables, keyboard focus, selectable rows with context. Maps always have list alternatives; floor controls support keyboards; safety information is never conveyed solely by color. SOS does not depend solely on a hold gesture; an explicit button + confirmation provides a fast path according to accessibility/site policy.

F01–F20 extend the coordinated screen set to 233 screens. Physical views include point/zone inventory, quality and freshness, accepted capability evidence, target review, approval, separate receipt/feedback timeline, UNKNOWN reconciliation, impairment/test/restoration and FAT/SAT acceptance, plus rescue-equipment assignment. FR-PHY-01–

FR-PHY-16 governs these behaviors. Do not enable an action from a stale projection; server/edge recheck. A local-only or unsupported action shows its reason and authorized manual workflow. No AI response can submit an actuator command or alter controller configuration.

12. Test architecture

Unit tests for state guards/rules/unit conversion. Contract tests for OpenAPI/events/adapters. Integration tests for outbox/inbox, workflow restart, DB tenant isolation, provider callbacks, and replay. E2E tests by actor and vi/en/ja, offline/reconnect, accessibility, and role changes. Performance tests for steady/burst/fan-out, soak, noisy neighbors, and media overload. Chaos tests for broker failure, DB failover, workflow-worker termination, provider timeouts, region partition, and clock skew. Security tests for BOLA, source spoofing, SSRF URL validation, upload malware fixtures, replay nonces, key revocation, and support break-glass. Separate hardware-in-loop tests for each actuator/drone vendor in an authorized environment.

Synthetic test data, deterministic clocks for deadlines, and mock providers represent accepted/delivered/unknown separately. Passing prototype navigation is not equivalent to passing production E2E. The acceptance matrix includes actor, trigger, expected evidence, and architecture owner so independent QA can design tests rather than merely mirror implementation.

Physical acceptance adds exact-configuration FAT and installed SAT. Test every enabled point/zone mapping and capability, negative cross-tenant/site/zone authorization, wrong revision, expired approval, duplicate event, missing feedback, and accepted-without-completion. Disconnect cloud, power off edge, restart gateway, isolate a partial point group, skew clocks and restore connectivity; verify local automatic protection continues and old intents do not actuate. Test hardware under an approved impairment plan with qualified local staff; simulator success is not authority for live discharge. FMEA assigns site owners and residual-risk decisions for wrong-zone actuation, lost feedback, supply failure, relay fault, test/actual confusion and compromised gateway. The detailed test/governance gates are in Standards and Deployment.

13. Key runbooks

Failure	Detection	Containment / recovery	Verify
Ingestion lag	oldest event age + p95	Reserve SOS lane; scale consumers; quarantine faulty source	No lost receipts, replay preserved IDs
Provider outage	failed/unknown attempts + callback age	Circuit break; alternate channel; operator visibility	Human ACK tracked separately; no duplicate flood
DB primary loss	health + write errors	Fence; promote verified replica; refresh service connections	Committed SOS present; no dual writer
Region unavailable	multi-vantage synthetic	Approved DR routing; epoch increase; outbox reconcile	Residency unchanged or explicitly approved; RPO/RTO measured
False detector storm	incident rate + operator feedback	Disable specific rule/model deployment; retain raw signals	Manual SOS unaffected; rollback version audited
Lost drone link	telemetry age	Expose unknown; local pilot/lost-link procedure	Receipt and physical state reconciled; no blind command replay
Compromised device	abnormal source behavior	Revoke cert; quarantine; rotate binding; coverage warning	Denied traffic; replacement device validated

Failure	Detection	Containment / recovery	Verify
Translation error	report/locale review	Withdraw affected template version; reviewed correction	Version lineage and recipients of correction visible
Cloud or edge loss at site	heartbeat/quality age, local panel observation	Mark remote integration unavailable; preserve local protection; buffer/reconcile supported observations	Independent automatic sequence verified; no stale write replay
Wrong zone or binding revision	point walkdown, digest mismatch	Suspend affected remote capability; retain local operation; correct/reaccept mapping	Point-to-point and negative cross-zone tests signed
Physical outcome unknown	accepted/sent without required feedback	UNKNOWN; qualified local verification; alternative response; no blind retry	Evidence-linked reconciliation with unchanged command history
Impairment or test overrun	planned_until, controller mode mismatch	Notify site responsible person; follow approved compensatory plan and restoration procedure	Local restored state, witness and signoff before closure
Rescue equipment link/console unavailable	operator report, supported telemetry age	Operator follows vendor lost-link/stop procedure; mission shows unknown/unavailable	Local physical condition and operator handback recorded

14. Delivery decomposition for the full build

Concurrent workstreams belong to the same complete scope: platform/IAM/regions; events/connectors/devices; incident/workflow/dispatch; Companion/Field; Command/admin; content/i18n; media/privacy/audit; observability/DR/security; hardware integration. Dependency order: contracts/domain invariants → platform foundations → vertical flows → complete role surfaces → integrations + resilience → independent acceptance. Administrative UI and error states are not omitted merely because they rarely appear in demos.

Definition of Ready: approved actors/permissions, state transitions, API/event schemas, error cases, translated copy, test fixtures. Definition of Done: code + migrations + contract tests + UI in three locales + accessible errors + dashboards/runbook + rollback + security/tenant tests + reviewer signoff. Production go-live requires domain/site authority validation; this document is a detailed review baseline, not acceptance signoff.

Full-build workstreams now explicitly include physical point/zone integration, OT DMZ/site edge, approved vendor gateways, capability/feedback contracts, impairment/test governance, FAT/SAT, and rescue-equipment operations. Definition of Ready for a physical write includes documented manufacturer permission, site cause-and-effect and observable completion; Definition of Done includes exact-configuration acceptance and a local restoration runbook. Unsupported actions remain visibly unavailable while the full designed scope is retained. The proposed operations/events above must be kept aligned with the separate JSON contracts during implementation.

Required appendices within the same baseline

Rescue speakers and voice navigation adds WF-21–22. Positioning, optimization, AI, and care profiles adds WF-23–27. Both documents are components of the full design, not optional features for a later phase. Screen specifications provides FRs for each screen ID.